

Lista 07 - Funções de Resumo

O teste do algoritmo desenvolvido foi feito através da ferramenta kulala.nvim, que permite a execução de requests pelo terminal, disponível em [Kulala.nvim | GitHub](#).

1. Chamada com sucesso dos dois consumidores

a. Primeiro consumidor com api-key “APP-01”

```
tmux 2:python3
# @attach-cookie-jar
GET http://localhost:8080/products ✓ 17.59 ms
1 X-API-Key: APP-01
2 X-Timestamp: 1778883569
3 # X-Timestamp: {{timestamp}}
4 X-Signature: HKD9ytfMaAKJmPCBREQ+evq7rvVKYfe93qne504=
5
6
all-products.http [+]
```

Body (B) Headers (H) All (A) Verbose (V) Script Output (O) Report (R) Help (?) ← [] →

Request: 7/7 Code: 0 Duration: 17.59 ms Time: May 15 08:39:40 PM
URL: GET http://localhost:8080/products Env: dev Status: 200 Assert: +
Buffer: requests/all-products.http::3 Name: GET http://localhost:8080/products

```
{
  "id": "bbbf84da-7b49-4bc7-a444-a8888a8c8ade",
  "name": "SSD",
  "description": "ssd 1tb sandisk",
  "cost_price": 480.0,
  "sell_price": 680.0
},
{
  "id": "a2d9d4c1-f7a9-4d5c-91e2-b7c8d3f6a801",
  "name": "Processador",
  "description": "amd ryzen 7 7800x3d",
  "cost_price": 1800.0,
  "sell_price": 2599.0
},
{
  "id": "c3f8a8e7-b5d4-4f1c-8a9e-2d7f4b0c8021",
  "name": "Placa de Video",
  "description": "nvidia rtx 4070 ti super 16gb",
  "cost_price": 4200.0,
  "sell_price": 5899.0
},
{
  "id": "d4a1b7c8-e9f2-4a5d-8c3e-1f6b7a9d0832",
  "name": "Fonte",
  "description": "fonte corsair 850w 80 plus gold",
  "cost_price": 350.0,
  "sell_price": 899.0
},
{
  "id": "e5b2c8d9-f1a3-4b0c-9d7e-2f1a8b0c8043",
  "name": "Placa-Mãe",
  "description": "placa-mãe asus tuf gaming b650a",
  "cost_price": 900.0,
  "sell_price": 1499.0
},
{
  "id": "f6c3d9e1-a2b4-5c7d-8e1f-3a2b9c1d0854",
  "name": "Water Cooler",
  "description": "water cooler deepcool 360mm rgb",
  "cost_price": 450.0,
  "sell_price": 799.0
},
{
  "id": "g7d4e1f2-a3b5-6c8d-9e2f-4a3b0c2d0865",
  "name": "Gabinete",
  "description": "gabinete mid tower com vidro temperado",
  "cost_price": 250.0,
  "sell_price": 350.0
}
```

all-products.http [+]

b. Segundo consumidor com api-key “APP-02”

```
tmux 2:python3
# @attach-cookie-jar
GET http://localhost:8080/products ✓ 18.46 ms
1 X-API-Key: APP-02
2 X-Timestamp: 1778889422
3 # X-Timestamp: {{timestamp}}
4 X-Signature: apfEz0ML9c3K6/BOYL8m4y7xQU2Y8/Eyo0DU1=
5
6
all-products.http [+]
```

Body (B) Headers (H) All (A) Verbose (V) Script Output (O) Report (R) Help (?) ← [] →

Request: 7/7 Code: 0 Duration: 18.46 ms Time: May 15 08:37:17 PM
URL: GET http://localhost:8080/products Env: dev Status: 200 Assert: +
Buffer: requests/all-products.http::3 Name: GET http://localhost:8080/products

```
{
  "id": "bbbf84da-7b49-4bc7-a444-a8888a8c8ade",
  "name": "SSD",
  "description": "ssd 1tb sandisk",
  "cost_price": 480.0,
  "sell_price": 680.0
},
{
  "id": "a2d9d4c1-f7a9-4d5c-91e2-b7c8d3f6a801",
  "name": "Processador",
  "description": "amd ryzen 7 7800x3d",
  "cost_price": 1800.0,
  "sell_price": 2599.0
},
{
  "id": "c3f8a8e7-b5d4-4f1c-8a9e-2d7f4b0c8021",
  "name": "Placa de Video",
  "description": "nvidia rtx 4070 ti super 16gb",
  "cost_price": 4200.0,
  "sell_price": 5899.0
},
{
  "id": "d4a1b7c8-e9f2-4a5d-8c3e-1f6b7a9d0832",
  "name": "Fonte",
  "description": "fonte corsair 850w 80 plus gold",
  "cost_price": 350.0,
  "sell_price": 899.0
},
{
  "id": "e5b2c8d9-f1a3-4b0c-9d7e-2f1a8b0c8043",
  "name": "Placa-Mãe",
  "description": "placa-mãe asus tuf gaming b650a",
  "cost_price": 900.0,
  "sell_price": 1499.0
},
{
  "id": "f6c3d9e1-a2b4-5c7d-8e1f-3a2b9c1d0854",
  "name": "Water Cooler",
  "description": "water cooler deepcool 360mm rgb",
  "cost_price": 450.0,
  "sell_price": 799.0
},
{
  "id": "g7d4e1f2-a3b5-6c8d-9e2f-4a3b0c2d0865",
  "name": "Gabinete",
  "description": "gabinete mid tower com vidro temperado",
  "cost_price": 250.0,
  "sell_price": 350.0
}
```

all-products.http [+]

2. Chamada sem sucesso – enviando mensagem não consistente com o código MAC

- Alteração do MAC em uma requisição do primeiro consumidor.
- MAC utilizado: j48DXp3WNeNmTau6IbJQB+kbGYI=
- MAC esperado: O4KKhDHLhqfGjcdj48DXp3WNeNmTau6IbJQB+kbGYI=

The screenshot shows a terminal window with a tmux session. The user is running a Python script that sends an HTTP GET request to `http://localhost:8080/products`. The request includes headers for `X-API-Key`, `X-Timestamp`, and `X-Signature`. The `X-Signature` is `j48DXp3WNeNmTau6IbJQB+kbGYI=`. The response from the server is a JSON object with a `detail` field: `"Assinatura inválida."`. The terminal output shows the request and response details, including the status code 401 and the assertion failure.

3. Chamada sem sucesso – enviando mensagem idêntica à chamada 1, mas enviada minutos depois (para demonstrar a solução que impede o replay attack)

Request para o 1º consumidor idêntica à da primeira questão, mas após 30 segundos

The screenshot shows a terminal window with a tmux session. The user is running a Python script that sends an HTTP GET request to `http://localhost:8080/products`. The request includes headers for `X-API-Key`, `X-Timestamp`, and `X-Signature`. The `X-Signature` is `KMD9ytfRaAkJmPCBREKQ+eVq7rVVKGYf93qne564=`. The response from the server is a JSON object with a `detail` field: `"Request fora do intervalo de tempo de 30s."`. The terminal output shows the request and response details, including the status code 401 and the assertion failure.